

# Binary Relation

沈威宇

September 26, 2026

# Contents

|       |                                                                                           |   |
|-------|-------------------------------------------------------------------------------------------|---|
| 1     | Binary Relation (二元關係) . . . . .                                                          | 1 |
| I     | Binary relation . . . . .                                                                 | 1 |
| i     | Binary relation or correspondence . . . . .                                               | 1 |
| ii    | Homogeneous relation or endorelation . . . . .                                            | 1 |
| iii   | Transitive relation . . . . .                                                             | 1 |
| iv    | Incomparability . . . . .                                                                 | 1 |
| v     | Important properties. . . . .                                                             | 1 |
| II    | Equivalence relation (等價關係) . . . . .                                                     | 2 |
| i     | Equivalence relation . . . . .                                                            | 2 |
| ii    | Equivalence class (等價類) . . . . .                                                         | 2 |
| iii   | Quotient set (商集). . . . .                                                                | 2 |
| III   | Order (序). . . . .                                                                        | 2 |
| i     | Ordered set (序集) . . . . .                                                                | 2 |
| ii    | Preorder (預序), quasiorder, or quasi-order . . . . .                                       | 2 |
| iii   | Total preorder (全預序), total quasiorder, or total quasi-order . . . . .                    | 3 |
| iv    | Partial order (偏序) . . . . .                                                              | 3 |
| v     | Upward closure . . . . .                                                                  | 3 |
| vi    | Downward closure . . . . .                                                                | 3 |
| vii   | Closed upward/upwards. . . . .                                                            | 3 |
| viii  | Closed downward/downwards. . . . .                                                        | 3 |
| ix    | Strict partial order, strict preorder, strict quasiorder, or strict quasi-order . . . . . | 3 |
| x     | Strict weak order . . . . .                                                               | 4 |
| xi    | Total order (全序) or linear order (線性順序) . . . . .                                         | 4 |
| xii   | Strict total order, strict linear order, or strict total preorder . . . . .               | 4 |
| xiii  | (Totally/linearly) ordered group (有序群) . . . . .                                          | 4 |
| xiv   | (Totally/linearly) ordered field (有序域) . . . . .                                          | 4 |
| xv    | Prewellorder (預良序) . . . . .                                                              | 5 |
| xvi   | Well order or well-order (良序). . . . .                                                    | 5 |
| xvii  | Strict well order or strict well-order . . . . .                                          | 5 |
| xviii | Dedekind-complete . . . . .                                                               | 5 |

|                                                                           |   |
|---------------------------------------------------------------------------|---|
| xix Order homomorphism or Order-preserving function . . . . .             | 5 |
| xx Order isomorphism . . . . .                                            | 5 |
| xxi Ordered by inclusion . . . . .                                        | 5 |
| xxii Cofinal (共尾的) or frequent. . . . .                                   | 6 |
| xxiii Upper Bound (上界) . . . . .                                          | 6 |
| xxiv Lower Bound (下界) . . . . .                                           | 6 |
| xxv Supremum, join, or Least Upper Bound (最小上界) . . . . .                 | 6 |
| xxvi Infimum, meet, or Greatest Lower Bound (最大下界) . . . . .              | 6 |
| xxvii Bounded (有界的) . . . . .                                             | 6 |
| xxviii Join-semilattice . . . . .                                         | 7 |
| xxix Meet-semilattice . . . . .                                           | 7 |
| xxx Lattice . . . . .                                                     | 7 |
| xxxi Directed set (有向集), directed preset (有向預序集), or filtered set . . . . | 7 |
| xxxii Product of directed sets . . . . .                                  | 7 |
| xxxiii Intervals (區間). . . . .                                            | 7 |

# 1 Binary Relation (二元關係)

## I Binary relation

### i Binary relation or correspondence

A binary relation or correspondence  $R$  over sets  $X$  and  $Y$  is a subset of  $X \times Y$ . The set  $X$  is called the domain or set of departure of  $R$ , and the set  $Y$  the codomain or set of destination of  $R$ . In order to specify the choices of the sets  $X$  and  $Y$ , some authors define a binary relation or correspondence as an ordered triple  $(X, Y, G)$ , where  $G$  is a subset of  $X \times Y$  called the graph of the binary relation. The statement  $(x, y) \in R$  reads " $x$  is  $R$ -related to  $y$ " and is denoted by  $xRy$ . The domain of definition or active domain of  $R$  is the set of all  $x \in X$  such that there exists  $y \in Y$  such that  $xRy$ . The codomain of definition, active codomain, image, or range of  $R$  is the set of all  $y \in Y$  such that there exists  $x \in X$  such that  $xRy$ . The field of  $R$  is the union of its domain of definition and its codomain of definition.

### ii Homogeneous relation or endorelation

A homogeneous relation or endorelation on a set  $X$  is a binary relation between  $X$  and itself, i.e. it is a subset of  $X^2$ , commonly phrased as "a (binary) relation on/over  $X$ ".

### iii Transitive relation

A homogeneous relation  $R$  on the set  $X$  is a transitive relation if

$$\forall a, b, c \in X : (aRb \wedge bRc) \implies aRc.$$

### iv Incomparability

For a homogeneous relation  $R$  on the set  $X$ , the homogeneous relation "incomparable with respect to  $R$ " is defined as " $a$  and  $b$  in  $X$  are incomparable with respect to  $R$  if and only if  $(\neg aRb) \wedge (\neg bRa)$ ".

### v Important properties

Some important properties that a homogeneous relation  $R$  over a set  $X$  may have include: for all  $a, b, c \in X$ ,

- Reflexivity:  $aRa$ ,
- Irreflexivity or anti-reflexive:  $\neg aRa$ ,
- Symmetry:  $aRb \implies bRa$ ,
- Asymmetry:  $aRb \implies \neg bRa$  (which implies irreflexivity),
- Antisymmetry:  $aRb \wedge bRa \implies a = b$ ,
- Transitivity:  $(aRb \wedge bRc) \implies aRc$ ,

- Transitivity of incomparability or comparability modulo equivalence: the homogeneous relation "incomparable with respect to  $R$ " is transitive, that is, an equivalence relation.
- Connectivity:  $a \neq b \implies (aRb \vee bRa)$  (which implies transitivity of incomparability),
- Strongly connectivity or total connectivity:  $aRb \vee bRa$  (which implies reflexivity and connectivity),
- Dense property:  $aRb \implies (\exists z \in X \text{ s.t. } aRz \wedge zRb)$ ,
- Well-founded property: for every non-empty subset  $S \subseteq X$ , there exists some  $m \in S$  such that  $mRs$  for all  $s \in S \setminus \{m\}$ .

## II Equivalence relation (等價關係)

### i Equivalence relation

An equivalence relation on a set  $X$  is a binary relation on  $X$  that is reflexive, symmetric, and transitive, often denoted as  $\cong$ ,  $=$ ,  $\equiv$ , or  $\sim$ .

### ii Equivalence class (等價類)

An equivalence class of an element  $a$  in a set  $X$  under an equivalence relation  $\cong$  on  $X$  is defined as

$$[a] = \{x \in X \mid a \cong x\}.$$

### iii Quotient set (商集)

The quotient set  $Y = X / \cong$  is the set of equivalence classes of elements of  $X$  by equivalence relation  $\cong$ .

## III Order (序)

### i Ordered set (序集)

An ordered set is an ordered pair  $P = (X, \leq)$  (or  $P = (X, <)$ ) consisting of a set  $X$ , called the ground set of  $P$ , and an order (序), aka ordering,  $\leq$  (or  $<$ ), in which an order is a transitive relation on  $X$ .

When the meaning is clear from context and there is no ambiguity about the order, the ground set itself can sometimes represents the whole ordered set.

### ii Preorder (預序), quasiorder, or quasi-order

A preorder, quasiorder, or quasi-ordering is a transitive relation  $\leq$  that is reflexive.

A set equipped with a preorder is called a preordered set, preset, quasiordered set, or quasi-ordered set.

### iii Total preorder (全預序), total quasiorder, or total quasi-order

A total preorder is a transitive relation  $\leq$  that is strongly connected.

A set equipped with a total preorder is called a totally preordered set, totally preset, totally quasiordered set, or totally quasi-ordered set.

### iv Partial order (偏序)

A partial order is a transitive relation  $\leq$  that is reflexive and antisymmetric.

A set equipped with a partial order is called a partially ordered set or poset.

### v Upward closure

Let  $A$  be a subset of a poset  $X$ , the upward closure of  $A$ , denoted as  $\uparrow A$ , is defined as:

$$\uparrow A := \{x \in X \mid \exists a \in A \text{ s.t. } a \leq x\}.$$

### vi Downward closure

Let  $A$  be a subset of a poset  $X$ , the downward closure of  $A$ , denoted as  $\downarrow A$ , is defined as:

$$\downarrow A := \{x \in X \mid \exists a \in A \text{ s.t. } a \geq x\}.$$

### vii Closed upward/upwards

A subset  $A$  of a poset  $X$  is called closed upward/upwards iff

$$(x \in X \wedge y \in A \wedge x \geq y) \implies (x \in A).$$

### viii Closed downward/downwards

A subset  $A$  of a poset  $X$  is called closed downward/downwards iff

$$(x \in X \wedge y \in A \wedge x \leq y) \implies (x \in A).$$

### ix Strict partial order, strict preorder, strict quasiorder, or strict quasi-order

A strict partial order, strict preorder, strict quasiorder, or strict quasi-order is a transitive relation  $<$  that is asymmetric.

A set equipped with a strict partial order is called a strictly partially ordered set, strictly preordered set, strictly preset, strictly quasiordered set, or strictly quasi-ordered set.

A strict partial order  $<$  can be induced from a preorder  $\leq$  with  $a < b \iff (a \leq b \wedge \neg(b \leq a))$ . A partial order  $\leq$  can be induced from a strict partial order  $<$  and an equivalence relation  $=$  with  $a \leq b \iff (a < b \vee a = b)$ .

## x Strict weak order

A strict weak order is a transitive relation  $<$  that is asymmetric and with transitivity of incomparability.

A strict weak order  $<$  can be induced from a preorder  $\leq$  with  $a < b \iff (a \leq b \wedge \neg(b \leq a))$ . A partial order  $\leq$  can be induced from a strict weak order  $<$  with  $a \leq b \iff (a < b \vee a = b)$  where  $=$  denotes the equivalence relation "incomparable with respect to  $<$ ".

## xi Total order (全序) or linear order (線性順序)

A total order or linear order is a transitive relation  $\leq$  that is strongly connected and antisymmetric.

A set equipped with a total order is called a totally ordered set or linearly ordered set.

## xii Strict total order, strict linear order, or strict total preorder

A strict total order, strict linear order, or strict total preorder is a transitive relation  $<$  that is asymmetric and connected.

A set equipped with a strict total order is called a strictly totally ordered set, strictly linearly ordered set, or strictly totally preordered set.

A strict total order  $<$  can be induced from a total preorder  $\leq$  with  $a < b \iff (a \leq b \wedge \neg(b \leq a))$ . A total order  $\leq$  can be induced from a strict total order  $<$  with  $a \leq b \iff (a < b \vee a = b)$  where  $=$  denotes the equivalence relation "incomparable with respect to  $<$ ".

## xiii (Totally/linearly) ordered group (有序群)

A group  $(G, *)$  together with a total order  $\leq$  on  $F$  is a

- left-(totally/linearly) ordered group if  $\leq$  is left-invariant, that is

$$\forall a, b, c \in G : a \leq b \implies c * a \leq c * b.$$

- right-(totally/linearly) ordered group if  $\leq$  is right-invariant, that is

$$\forall a, b, c \in G : a \leq b \implies a * c \leq b * c.$$

- bi-(totally/linearly) ordered group if  $\leq$  is bi-invariant, that is it is both left- and right-invariant.

If  $*$  is commutative, then left-ordered, right-ordered, and bi-ordered groups are the same, called (totally/linearly) ordered group.

## xiv (Totally/linearly) ordered field (有序域)

A field  $(F, +, \cdot)$  together with a total order  $\leq$  on  $F$  is a (totally/linearly) ordered field if the order satisfies the following properties for all  $a, b, c \in F$ :

$$a \leq b \implies a + c \leq b + c,$$

$$0 \leq a \wedge 0 \leq b \implies 0 \leq a \cdot b.$$

Elements  $a \in F$  with  $a > 0$  are called positive;  $a \in F$  with  $a < 0$  are called negative.

#### **xv Prewellorder (預良序)**

A prewellorder is a transitive relation  $\leq$  that is strongly connected and well-founded.

A set equipped with a prewellorder is called a prewellordered set.

#### **xvi Well order or well-order (良序)**

A well order or well-order is a transitive relation  $\leq$  that is strongly connected, antisymmetric, and well-founded.

A set equipped with a well-order is called a well ordered set, well-ordered set, or woset.

#### **xvii Strict well order or strict well-order**

A strict well order or strict well-order is a transitive relation  $<$  that is asymmetric, connected, and well-founded.

A set equipped with a strict well order is called a strictly well ordered set or strictly well-ordered set.

A strict well order  $<$  can be induced from a prewellorder  $\leq$  with  $a < b \iff (a \leq b \wedge \neg(b \leq a))$ . A well order  $\leq$  can be induced from a strict well order  $<$  with  $a \leq b \iff (a < b \vee a = b)$  where  $=$  denotes the equivalence relation "incomparable with respect to  $<$ ".

#### **xviii Dedekind-complete**

A set  $X$  equipped with a total order  $\leq$  is Dedekind-complete if every nonempty subset  $S$  of  $X$  with an upper bound in  $X$  has a least upper bound (aka supremum) in  $X$ .

#### **xix Order homomorphism or Order-preserving function**

Given two posets  $(S, \leq_S)$  and  $(T, \leq_T)$ , an order homomorphism or order-preserving function from  $(S, \leq_S)$  to  $(T, \leq_T)$  is a function  $f$  from  $S$  to  $T$  with the property that, for every  $x, y \in S$ ,  $x \leq_S y$  implies  $f(x) \leq_T f(y)$ .

#### **xx Order isomorphism**

Given two posets  $(S, \leq_S)$  and  $(T, \leq_T)$ , an order isomorphism from  $(S, \leq_S)$  to  $(T, \leq_T)$  is a bijective function  $f$  from  $S$  to  $T$  with the property that, for every  $x, y \in S$ ,  $x \leq_S y$  if and only if  $f(x) \leq_T f(y)$ .

Two sets are called to be order-isomorphic if there exists an order isomorphism between them.

#### **xxi Ordered by inclusion**

A family of sets is called to be ordered by inclusion if it is a poset such that for any two sets  $A, B$  in it,  $A \leq B \iff A \subseteq B$ .



## xxii Cofinal (共尾的) or frequent

A subset  $B \subseteq A$  of a preordered set  $(A, \leq)$  is said to be cofinal or frequent in  $A$  if for every  $a \in A$ , there exists a  $b \in B$  such that  $a \leq b$ .

## xxiii Upper Bound (上界)

$M$  is an upper bound of a preset  $A$  if  $a \leq M$  for every  $a \in A$ . We say a preset is bounded(-)above by  $M$  if it has an upper bound  $M$ .

$M$  is an upper bound of a function  $f$  of which the range is a preset if  $M$  is an upper bound of its range. We say a function is bounded-above by  $M$  if it has an upper bound  $M$ .

$M$  is an upper bound of a function  $f$  of which the range is a preset on a subset  $I$  of its domain if  $M$  is an upper bound of  $f(I)$ . We say a function is bounded-above by  $M$  on a subset  $I$  of its domain if it has an upper bound  $M$  on  $I$ .

## xxiv Lower Bound (下界)

$m$  is a lower bound of a preset  $A$  if  $a \geq m$  for every  $a \in A$ . We say a preset is bounded(-)below by  $m$  if it has a lower bound  $m$ .

$m$  is a lower bound of a function  $f$  of which the range is a preset if  $m$  is a lower bound of its range. We say a function is bounded-below by  $m$  if it has a lower bound  $m$ .

$m$  is a lower bound of a function  $f$  of which the range is a preset on a subset  $I$  of its domain if  $m$  is a lower bound of  $f(I)$ . We say a function is bounded-below by  $m$  on a subset  $I$  of its domain if it has a lower bound  $m$  on  $I$ .

## xxv Supremum, join, or Least Upper Bound (最小上界)

The supremum, join, or least upper bound of a preset  $A$ , denoted as  $\sup A$ , is defined as  $M_0$  in the set  $\mathcal{M}$  of all upper bounds of  $A$  such that  $\forall M \in \mathcal{M} : M_0 \leq M$ .

The supremum or least upper bound of a function  $f$  of which the range is a preset, denoted as  $\sup_x f(x)$ , is defined as  $M_0$  in the set  $\mathcal{M}$  of all upper bounds of  $f$  such that  $\forall M \in \mathcal{M} : M_0 \leq M$ .

## xxvi Infimum, meet, or Greatest Lower Bound (最大下界)

The infimum, meet, or greatest lower bound of a preset  $A$ , denoted as  $\inf A$ , is defined as  $m_0$  in the set  $\mathcal{M}$  of all lower bounds of  $A$  such that  $\forall m \in \mathcal{M} : m_0 \geq m$ .

The infimum or greatest lower bound of a function  $f$  of which the range is a preset, denoted as  $\inf_x f(x)$ , is defined as  $m_0$  in the set  $\mathcal{M}$  of all lower bounds of  $f$  such that  $\forall m \in \mathcal{M} : m_0 \geq m$ .

## xxvii Bounded (有界的)

We say a preset or a function of which the range is a preset is bounded if it is bounded-above and bounded-below.

We say a function of which the range is a preset is bounded on a subset  $I$  of its domain if it is bounded-above and bounded-below on  $I$ .

We say a function  $f$  of which the range is a preset is unbounded at a point  $a$  in its domain if  $f$  is unbounded on all neighborhood of  $a$ .

### xxviii Join-semilattice

A join-semilattice is a partially ordered set such that every two-element  $\{a, b\}$  subset has a join.

### xxix Meet-semilattice

A meet-semilattice is a partially ordered set such that every two-element  $\{a, b\}$  subset has a meet.

### xxx Lattice

A lattice is a partially ordered set such that every two-element  $\{a, b\}$  subset has a join and a meet.

### xxxi Directed set (有向集), directed preset (有向預序集), or filtered set

A directed set is a preset in which every finite subset has an upper bound.

### xxxii Product of directed sets

Suppose  $(A_i, \leq_i)_{i \in I}$  is an indexed family of directed sets.  $A = \prod_{i \in I} A_i$  is a directed set equipped with coordinatewise order  $(a_i)_{i \in I} \leq (b_i)_{i \in I} \equiv (a_i \leq_i b_i)_{i \in I}$ .

### xxxiii Intervals (區間)

Unless otherwise specified, an interval defined for  $\mathbb{R}$ .

For a partially ordered set  $(P, \leq)$  and  $a, b \in P$ ,

- a closed interval (閉區間)  $[a, b]$  is defined as

$$[a, b] := \{x \in P \mid a \leq x \leq b\};$$

- an left-closed upward interval  $[a, \infty)$  is defined as

$$[a, \infty) := \{x \in P \mid a \leq x\};$$

- a right-closed downward interval  $(-\infty, b]$  is defined as

$$(-\infty, b] := \{x \in P \mid x \leq b\};$$

- the open interval  $(-\infty, \infty)$  is defined as

$$(-\infty, \infty) := P.$$

For a strictly partially ordered set  $(P, <)$  and  $a, b \in P$ ,

- an open interval (開區間)  $(a, b)$  is defined as

$$(a, b) := \{x \in P \mid a < x < b\};$$

- a right open interval (右開區間)  $[a, b)$  is defined as

$$[a, b) := \{x \in P \mid a \leq x < b\};$$

- a left open interval (左開區間)  $(a, b]$  is defined as

$$(a, b] := \{x \in P \mid a < x \leq b\};$$

- an left-open upward interval  $(a, \infty)$  is defined as

$$(a, \infty) := \{x \in P \mid a < x\};$$

- a right-open downward interval  $(-\infty, b)$  is defined as

$$(-\infty, b) := \{x \in P \mid x < b\}.$$

Right open intervals and left open intervals are called half open intervals (半開區間). Intervals with at endpoint  $\infty$  or  $-\infty$  are called infinite intervals (無限區間) or unbounded intervals. All intervals that are not infinite intervals are called bounded intervals. The length of a bounded interval is the difference of the two endpoints of it.